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Corrigendum to "Microplastics generated from a biodegradable plastic in freshwater and seawater" [Water Research, 198(2021), 117123]☆

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In this study, a large number of particles of various shape and size were observed in both freshwater and seawater during the ageing of both pristine and UV-aged films, made of biodegradable poly(butylene adipate-co-terephthalate) (PBAT) and non-biodegradable low-density polyethylene (LDPE), for up to 10 weeks. It was concluded that the formation of microplastics occurred more readily from PBAT than from LDPE in the aquatic environments.

The observed microplastics were characterized using optical and scanning electron microscopy. It was concluded that these particles were microplastics formed during PBAT and LDPE degradation, based on two assumptions: (i) that the Milli-Q water used was ultrapure and did not support microbial growth, and (ii) that the observed fibrous particles exhibited a white color similar to that of the PBAT films. Following feedback on the results and additional characterization (see below), we realized that our initial assumptions were incorrect and that some types of the observed particles were most likely of microbial origin that developed during the experiments.

The authors regret that, due to the absence of chemical identification, certain particles were incorrectly classified as microplastics. Consequently, the present study cannot substantiate the conclusion that biodegradable plastics release more microplastics than non-biodegradable plastics under the tested conditions.

We managed to recover the glass slides on which water droplets from the exposure experiments had been placed and evaporated. Although the slides were approximately five years old and the material collected on these may have undergone some degree of degradation, attempts were still made to analyze the composition of the observed particles on the glass sides using an IR microscope (LUMOS, Bruker). The original samples were prepared by pipetting 25 μL of water taken from the flasks into wells on the glass slides designed for volumes up to 30 μL, followed by drying in a cleanroom at room temperature for one day. The glass slides were then stored in a closed, but not air-tight or moisture free, box at ambient conditions. The glass slide was mounted on the LUMOS microscope sample holder using double-sided adhesive tape. The ATR mode was employed with a measurement area of 20×20 μm^2 and 64 scans per spectrum. The ATR crystal was wiped with ethanol-moistened tissue after each measurement to prevent cross-contamination.

Figs. 1 and 2 show representative optical microscopy images of the samples observed using the IR microscope, along with the corresponding IR spectra of the fibrous and particulate particles measured at the marked locations. Fig. 3 further compares the spectra of the observed particles with those of pure PBAT and glass slides. More than ten particles from the Milli-Q water and artificial seawater samples, in which the PBAT films had been exposed for 10 weeks, were analyzed, all exhibiting similar spectral characteristics. The spectra display a broad band around 3300 cm^{-1}, attributed to O–H stretching, a peak at 1630 cm^{-1}, and a peak at 1110 (for particulate particles) and 1078 cm^{-1} (for the fibrous particles). Many characteristic PBAT peaks, e.g., the main carbonyl peak at 1710 cm^{-1}, were absent. This indicates that the examined fibers and particles were not PBAT. The observed bands near ~1100 cm^{-1} may be attributed to C–O stretching vibrations that can arise from polysaccharides or nucleic acids associated with microbial material. However, a definitive assignment to microbial origin could not be made, as characteristic amide I and amide II bands (~1650 and ~1550 cm^{-1}) and aliphatic lipid bands (~2800–3000 cm^{-1}), typically associated with microbial biomass, were not clearly observed (Dukor, 2001). The reason for the difficulty in identifying the origin of the particles was the strong peaks of water that may overlap the absorbance from other components. The bands associated with moisture or residual water in the particles are observed at 3000–3500 and 1630 cm^{-1}, corresponding to the O–H stretching and H–O–H bending vibrations of water (Mojet et al., 2010). An gradual upturn of the IR absorbance from

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